

Mutual Exclusion of Local Divisibility Constraints in the Collatz Cycle Equation

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Abstract

We provide computational evidence that the standard heuristic of Crandall (1978) for the cyclic component of the Collatz problem fails in a structured way. For all 22 admissible parity patterns (x, y) enumerated exhaustively (total composition count $\sum |\Sigma| \approx 1.77 \times 10^9$), the only solutions to the integrality condition $(2^x - 3^y) \mid C(y, \vec{\sigma})$ correspond to the trivial cycle $a_i = 2$ at $x = 2y$. Crandall's heuristic predicts an aggregate of 7.21 non-trivial cycle solutions; observed: 0 ($p \approx 7 \times 10^{-4}$). The deviation is structurally explained by mutual exclusion of local divisibility events: when $2^x - 3^y$ has multiple prime factors, the marginal events $\{p_i \mid C\}$ each occur at their expected Crandall rate, but their joint occurrence is empirically empty. Across 5 cases with composite $2^x - 3^y$, the independence prediction sums to 8.86 joint occurrences; observed: 0 ($p \approx 1.4 \times 10^{-4}$). The object studied is the *Collatz corona* of Belaga–Mignotte (1998); the explicit mutual-exclusion observation does not appear, to the author's knowledge, in the prior literature surveyed.

1 Setup and prior work

The accelerated Collatz map $T'(n) = (3n + 1)/2^{v_2(3n+1)}$ on odd integers has cyclic orbits parameterized by sequences $0 = \sigma_0 < \sigma_1 < \dots < \sigma_{y-1} < x$ via the cycle equation [?]

$$n_0(2^x - 3^y) = C(y, \vec{\sigma}) := \sum_{k=0}^{y-1} 3^{y-1-k} 2^{\sigma_k}. \quad (1)$$

A non-trivial cycle requires $2^x > 3^y$ and $(2^x - 3^y) \mid C(y, \vec{\sigma})$. With composition $a_k = \sigma_k - \sigma_{k-1}$ and $a_y = x - \sigma_{y-1}$, all $a_k \geq 1$ and $\sum a_k = x$. Write $\Sigma(x, y)$ for the set of admissible patterns; $|\Sigma(x, y)| = \binom{x-1}{y-1}$.

Belaga and Mignotte [?] introduced this framework explicitly: the integer $2^x - 3^y$ is a *Collatz number*, and the set $\{C(y, \vec{\sigma}) : \vec{\sigma} \in \Sigma(x, y)\}$ is its *Collatz corona*. The trivial cycle $n_0 = 1$ corresponds to $a_k = 2$ for all k , requiring $x = 2y$.

The standard heuristic [?] models $C(y, \vec{\sigma}) \bmod (2^x - 3^y)$ as uniformly distributed, giving expected cycle count $E_{\text{Cr}}(x, y) = \binom{x-1}{y-1} / (2^x - 3^y)$ per pair.

Existing rigorous results on cycles approach the problem via *linear forms in logarithms* (Baker–Wüstholz, Laurent–Mignotte–Nesterenko). Steiner [?] excluded 1-cycles. Simons and de Weger [?] proved any non-trivial cycle has at least $m \geq 68$ local minima; Hercher [?] raised this to $m \geq 92$. Belaga [?] gave polynomial upper bounds on the *number* of $(3x + d)$ -cycles of given odd-length. Belaga and Mignotte [?] computed primitive cycles for $1 \leq d \leq 19,999$.

Recently, Dhiman and Pandey [?] showed that the divisibility predicate $\mathcal{D}_y = \{(x, C) \in \mathbb{N}^2 : (2^x - 3^y) \mid C\}$ is not Presburger-definable.

2 Empirical observations

Aggregate rate. We enumerated $\Sigma(x, y)$ exhaustively for 22 pairs with 1.77×10^9 total compositions; see Table ???. Across non-trivial pairs $(x \neq 2y)$,

$$\sum_{(x,y) \neq (2y,y)} E_{Cr}(x, y) \approx 7.21.$$

Observed non-trivial solutions: 0. Modeling cases as independent Poisson, $P(\text{count} = 0) \approx e^{-7.21} \approx 7.4 \times 10^{-4}$.

Table 1: Selected enumerated pairs. Subset of 22 cases. “ E_{Cr} ” is the Crandall expected zero count.

y	x	$ \Sigma $	$2^x - 3^y$	E_{Cr}	non-triv. zeros
5	8	35	13	2.69	0
9	15	3,003	13,085	0.23	0
10	16	5,005	6,487	0.77	0
12	20	75,582	517,135	0.15	0
15	24	817,190	2,428,309	0.34	0
16	26	3,268,760	24,062,143	0.14	0
17	27	5,311,735	5,077,565	1.05	0
19	31	86,493,225	985,222,181	0.09	0
21	34	573,166,440	6,719,515,981	0.09	0
22	35	927,983,760	2,978,678,759	0.31	0
<i>aggregate $\sum E_{Cr}$ over all 22 enumerated non-trivial pairs:</i>				7.21	0

Mechanism: marginal vs. joint. Define $A_p := \{\vec{\sigma} \in \Sigma(x, y) : p \mid C(y, \vec{\sigma})\}$. For $(x, y) = (24, 15)$, $2^x - 3^y = 13 \cdot 186,793$:

$$|A_{13}| = 62,775, \quad |A_{186793}| = 50, \quad |A_{13} \cap A_{186793}| = 0.$$

Independence prediction: $|A_{13}| \cdot |A_{186793}| / |\Sigma| \approx 3.84$.

Proposition 1 (Mutual exclusion, observed). *Across the 5 enumerated pairs (15, 9), (16, 10), (20, 12), (24, 15), (26, 16) with composite $2^x - 3^y$, the aggregate independence prediction is 8.86. Observed joint occurrences: 0. Modeled as Poisson, $P \approx e^{-8.86} \approx 1.4 \times 10^{-4}$.*

3 Conjecture

Conjecture 1. *For every $(x, y) \in \mathbb{N}^2$ with $2^x > 3^y$ and $(x, y) \neq (2y, y)$,*

$$\{C(y, \vec{\sigma}) : \vec{\sigma} \in \Sigma(x, y)\} \cap (2^x - 3^y)\mathbb{Z} = \emptyset.$$

If true, Conjecture ?? implies non-existence of non-trivial Collatz cycles by purely structural means, complementing the asymptotic length bounds of Steiner–Simons–de Weger–Hercher.

4 Position within the literature

The Collatz number / Collatz corona framework dates from Belaga–Mignotte 1998 [?]. Existing rigorous results follow two complementary strategies:

Diophantine bounds via linear forms in logarithms [?, ?, ?, ?]: lower bounds on cycle length m via Baker-type estimates on $|x \log 2 - y \log 3|$.

Combinatorial enumeration [?, ?, ?]: upper bounds on the number of $(3x + d)$ -cycles of given odd-length, and explicit listing of primitive cycles for $d \geq 1$.

The present note proposes a third angle: *mutual exclusion* between distinct prime divisors of the Collatz number, observed empirically as a deviation from Crandall’s heuristic. To the author’s knowledge, this specific phenomenon has not been stated in the prior literature surveyed (Belaga–Mignotte 1998, 2000, 2006; Belaga 2003; Simons–de Weger 2005; Simons 2008; Holden 2014; Hercher 2018, 2022; Dhiman–Pandey 2026). The author has not, however, performed an exhaustive bibliographic survey, and any precedent will be acknowledged in subsequent versions.

A proof of Conjecture ?? would presumably require analytic input beyond combinatorial counting – e.g. subspace-theorem-type bounds on S -unit sums with $S = \{2, 3\}$ [?, ?] – since the relevant deviation lives in the conjunction of local arithmetic constraints, which Dhiman–Pandey [?] show cannot be captured within Presburger arithmetic.

5 Limitations

1. Enumeration is bounded by $|\Sigma| \leq 10^9$ on a single machine; convergent pairs (x, y) of $\log_2 3$ beyond $(65, 41)$ are inaccessible by exhaustive search.
2. The mutual exclusion mechanism is verified explicitly across 5 cases with composite $2^x - 3^y$. Verification across additional composite-diff cases is the immediate next step.
3. The mechanism behind marginal-versus-joint discrepancy is empirically confirmed but not derived from a stated p -adic principle.
4. Conjecture ?? concerns only the cyclic component of Collatz. The divergent component is independent and unaffected.
5. The independence-of-cases assumption underlying our Poisson p -values is an approximation.

Code and data. Available at <https://github.com/<USER>/collatz-mutual-exclusion>.

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